

Dependency and Scope: On the Translation Between Lemmon and Fitch Proofs

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Abstract

Lemmon-style and Fitch-style natural deduction are usually treated as notational variants of one another. They are, but the sense in which they are is more delicate than that description suggests, and the delicacy is instructive. A Lemmon line carries an explicit and exact record of the assumptions it rests on; a Fitch line carries no such record, and its dependencies are bounded, but not determined, by the subproofs enclosing it. We show that translation from Fitch to Lemmon is total and canonical, that the dependency column of a Lemmon proof is therefore redundant, and that translation in the other direction is total as well—but not *positionally*: there are Lemmon proofs whose lines cannot be carried over one for one, in order, to any Fitch proof. We exhibit two such proofs, identify the two obstructions responsible, and show that both obstructions are artefacts of linearisation rather than facts about the two systems: they cannot even be stated of the underlying derivation trees. All the results were implemented and machine-checked against a corpus of 78 proofs together with sixteen proofs photographed from student work.

1 Introduction

Anyone who has taught both notations knows that a Lemmon proof and a Fitch proof of the same argument look like transcriptions of one another. In the Lemmon style, each line is annotated with the numbers of the assumptions it depends on, and a rule such as conditional proof discharges an assumption by removing its number from the annotation. In the Fitch style, the same information is carried geometrically: an assumption opens a subproof, drawn as a vertical scope line, and discharging it closes the subproof. The two conventions are so evidently doing the same work that the translation between them is generally left as an exercise, or not mentioned.

It is not quite an exercise. The purpose of this paper is to say exactly what the translation is, where it is easy, where it is not, and what the difficulty reveals about the relationship between the two notations.

The central observation is simple enough to state here. In a Lemmon proof, the dependency set attached to a line is *exact*: it is the set of assumptions the line actually rests on. In a Fitch proof, the corresponding information is the set of assumptions of the enclosing subproofs, and this is merely an *upper bound*. A Fitch line may perfectly well sit inside a subproof whose assumption it never uses. Every asymmetry we discuss below follows from this one.

A second observation, less obvious, does most of the work in Section 4. In a Lemmon proof, the textual position of a line carries no semantic content. Lines may be permuted freely,

subject only to the requirement that a line be written after the lines it cites. In a Fitch proof, position is semantic: where a line sits determines what may see it, and a line inside a subproof becomes unavailable once that subproof closes. The translation from Lemmon to Fitch must therefore invent positional information that the source does not contain, and it is possible for the source to place demands that no such invention can satisfy.

We work throughout with the rules of *How Logic Works* [1], a Lemmon-style system with twenty-one rules, and with a Fitch system chosen to match it rule for rule. Section 2 fixes notation. Section 3 treats the easy direction and draws from it the redundancy of the dependency column. Section 4 exhibits the two obstructions to a positional translation in the other direction. Section 5 shows that both dissolve when one passes through derivation trees, and that translation is consequently total. Section 6 states two questions we have not settled. Section 7 reports the machine verification, and Section 8 draws the moral.

2 The two systems

Definition 1. A *Lemmon proof* is a finite sequence of lines. A line is a quadruple $\langle \Gamma, n, \varphi, j \rangle$ where n is the line’s number, φ its formula, j its justification, and Γ a set of line numbers—the *dependency set*. A justification is either A (assumption) or a rule together with the numbers of the lines it cites, all of which must be smaller than n .

The rules fall into two groups. Most of them—modus ponens, conjunction introduction, universal elimination—simply pool the dependency sets of the lines they cite. Four of them discharge: conditional proof and reductio each cite an assumption line together with a line derived from it, and disjunction and existential elimination each cite one or two such pairs. For these, the discharged assumption is removed from the pooled set. Thus in

1	(1) P	A
2	(2) Q	A
1,2	(3) $P \wedge Q$	1,2 $\wedge$ I
1	(4) $Q \rightarrow (P \wedge Q)$	2,3 CP

line (4) discharges assumption (2), which accordingly leaves the dependency set.

Definition 2. A *Fitch proof* is a finite sequence of *items*, where an item is either a line—a number, a formula and a justification—or a *subproof*, consisting of an assumption line together with a further sequence of items. The *scope* of a line is the set of assumptions of the subproofs containing it. A line may cite another only if the cited line lies within its scope, that is, only if every subproof containing the cited line also contains the citing line.

The same proof in Fitch:

1	P	Premise
2	Q	Assume
3	$P \wedge Q$	$\wedge$ I 1,2
4	$Q \rightarrow (P \wedge Q)$	CP 2–3

Two conventions deserve comment. First, an assumption never discharged is written in Fitch as a premise at the outermost level, so the choice between “premise” and “assumption”

is determined by whether the line is discharged later; nothing in the Lemmon proof marks the difference locally. Second, we take universal introduction to carry a side condition on the name generalised upon, rather than to open a subproof with a flagged constant. Both conventions are found in the literature; the first agrees with the condition already imposed in the Lemmon system, and choosing it lets universal introduction translate one line to one line.

3 From Fitch to Lemmon

This direction is unproblematic, and it is worth seeing why, because the reason is exactly what fails in the other direction.

Definition 3. Let F be a Fitch proof. Define $\delta(F)$, the Lemmon proof induced by F , by flattening F into a sequence of lines in order and assigning to each line n the set $\Gamma(n)$ given by: $\Gamma(n) = \{n\}$ if n is an assumption or premise; $\Gamma(n) = \bigcup_m \Gamma(m)$ for m ranging over the lines cited, if the rule at n discharges nothing; and for a discharging rule, the same union with the discharged assumptions removed.

Theorem 4. δ is total: for every Fitch proof F , $\delta(F)$ is a Lemmon proof, and F is correct if and only if $\delta(F)$ is.

Proof sketch. The definition of Γ is a recursion on the position of the line, which terminates because a line may cite only earlier lines. That the resulting annotation satisfies the Lemmon conditions on each rule is checked rule by rule; the discharging rules are the only ones where anything happens, and for those the Fitch scope condition guarantees that the discharged assumption is available to be removed. $\square$

Proposition 5. For each line n of F , the set $\Gamma(n)$ assigned by δ is contained in the scope of n , and the containment may be proper.

The containment may be proper precisely when a line sits inside a subproof whose assumption it does not use. This has a consequence worth recording.

Corollary 6. δ is not injective. Two Fitch proofs that differ only in the placement of a line within subproofs it does not depend on have the same image.

Consequently the round trip from Fitch through Lemmon and back is a normalisation rather than an identity: it returns a proof whose subproofs are no wider than they need to be.

The next observation is elementary but, we think, has not been stated. Because the Lemmon rules determine the dependency set of each line exactly—a system that permitted a line to cite more assumptions than it uses would not satisfy Proposition 5—the dependency column is not independent data.

Proposition 7. Let L be a Lemmon proof and let L^- be the result of deleting its dependency column. Then L is recoverable from L^- .

Proof. The recursion in the definition of δ makes no use of the dependency column of the source; it uses only the justifications. Applying it to L^- returns L . $\square$

Remark 8. The pedagogical function of the dependency column is thus not to record information the reader could not otherwise obtain, but to require the student to compute it. This is a defensible reason for a notation to carry redundant data, and it is worth distinguishing from the reason a notation might carry data that is genuinely needed.

4 Two obstructions in the other direction

Definition 9. A translation from Lemmon to Fitch is *positional* if it preserves the sequence of lines: the n th line of the Lemmon proof becomes the n th line of the Fitch proof, with the same formula and the corresponding rule.

Positional translation is what one wants, when it is available. It preserves line numbers, so a student may read the two proofs side by side; and it certifies that the two proofs are the same proof in two notations, rather than two proofs of the same thing.

Theorem 10. *There are correct Lemmon proofs with no positional Fitch image.*

We give two, illustrating two independent obstructions.

	1	(1) P	A
	2	(2) Q	A
Example 11 (Discharge order).	1,2	(3) $P \wedge Q$	1,2 $\wedge$ I
	2	(4) $P \rightarrow (P \wedge Q)$	1,3 CP
		(5) $Q \rightarrow (P \rightarrow (P \wedge Q))$	2,4 CP

Assumption (1) is made first and discharged first. In a positional Fitch image, the subproof opened by (1) would have to contain the subproof opened by (2), since (1) is opened first and subproofs are properly nested; but a subproof may be closed only when every subproof it contains has been closed, and (4) closes (1) while (2) remains open. Fitch permits discharge only in last-in-first-out order, and Lemmon imposes no such restriction.

	1	(1) P	A
	2	(2) Q	A
Example 12 (Positional trapping).	1	(3) $P \vee R$	1 $\vee$ I
	2	(4) $Q \wedge Q$	2,2 $\wedge$ I
		(5) $Q \rightarrow (Q \wedge Q)$	2,4 CP
	1	(6) $(P \vee R) \wedge (Q \rightarrow (Q \wedge Q))$	3,5 $\wedge$ I

Here the discharge order is unobjectionable. The difficulty is that line (3) is written between assumption (2) and its discharge at (5), and so in any positional image lies inside the subproof opened by (2). It does not depend on (2)—its dependency set is $\{1\}$ —but that is of no help: when the subproof closes at (5), line (3) passes out of scope, and line (6) cites it. In the Lemmon proof nothing goes out of scope at all. Discharging an assumption changes the dependency set of the new line; it does not affect the availability of any earlier line.

Example 12 is the more instructive of the two, and it isolates the second of our opening observations. The Lemmon proof does not say that line (3) is “inside” anything. It is written where it is written, and that is a fact about the page, not about the derivation. A positional translation is obliged to read the page as though position meant something, and here it does not.

Remark 13. Both examples can be repaired by permuting lines. Swapping (1) and (2) in Example 11 yields a proof in which the discharges nest; moving (3) before (2) in Example 12 lifts it out of the subproof. Whether permutation always suffices is Conjecture 16 below.

5 Derivation trees, and totality

Both obstructions concern the linear order of lines. It is therefore natural to ask what becomes of them when the linear order is discarded.

Definition 14. A *derivation* is a finite tree. Leaves are assumptions, labelled with an identifier; an internal node is labelled with a formula and a rule, and a node whose rule discharges is labelled additionally with the identifier of the assumption it discharges, which must occur among the leaves of its subtree.

Every Lemmon proof determines a derivation, by unfolding: the derivation of the last line is built by recursion, replacing each cited line number by the derivation of that line. A line cited twice is thereby copied. Every derivation in turn determines a Fitch proof, by a traversal that emits the subderivations a rule requires and then the line applying it, opening a subproof for each discharge.

Theorem 15. *Every correct Lemmon proof has a Fitch image with the same conclusion and premises drawn from among its own.*

Proof sketch. Unfold and traverse, as just described. Neither obstruction can arise. A derivation has no linear order, so there is no discharge order to violate: a node that discharges an assumption closes exactly the subproof its own subtree opened, and nesting is automatic. And no node is ever cited from a sibling branch, since each occurrence of a line is a distinct subtree, so no line can be stranded by the closing of a subproof it does not belong to. $\square$

The two obstructions of Section 4 are therefore not facts about Lemmon and Fitch. They are facts about *linearisation*. Both notations are ways of writing a derivation down in a line; the difficulties arise only in the flattening, and the two notations flatten differently.

The construction has a cost. A line cited twice is derived twice, and in the worst case the unfolding is exponential in the size of the source. There is also a small point of rule bookkeeping: if a subproof's conclusion is a line from outside it—as in deriving $Q \rightarrow P$ from the premise P —then the subproof would contain nothing at all, and Fitch requires a reiteration step to bring the line in. Lemmon has no such rule and needs none, and the reiterated line is a tautological consequence of the line it repeats, so nothing is added to the system.

6 Two open questions

Conjecture 16. *Every correct Lemmon proof can be permuted—preserving the requirement that a line follow the lines it cites—into one with a positional Fitch image.*

Conjecture 17. *No translation requires duplication. That is, every correct Lemmon proof has a Fitch image in which each line of the source occurs exactly once.*

Conjecture 17 is suggested by the following consideration. If line L is cited by line M under a non-discharging rule, then $\Gamma(L) \subseteq \Gamma(M)$. Placing every line at the depth of the innermost assumption in its dependency set therefore places L no deeper than M , and hence within M 's scope. What this argument does not establish is that the resulting family of subproofs is laminar, which is what a Fitch proof requires, and we have not been able to settle whether it must be.

Some evidence bears on both conjectures. Examples 11 and 12 translate through derivation trees to Fitch proofs of five and six lines respectively—exactly the lengths of the originals. No line was duplicated in either case. This is what one would expect if Conjecture 17 holds and the unfolding is simply doing more work than necessary.

7 Machine verification

The translations in both directions were implemented in Haskell against the proof checker for *How Logic Works*, and the following were verified mechanically.

Every proof in a regression corpus of 78 cases was translated, where valid, from Lemmon to Fitch and back. Twenty-five admitted a positional translation, and for each the round trip returned the original proof line for line, *including its dependency sets*—an instance of Proposition 7, since the sets were recomputed and not copied. The two proofs of Section 4 were translated through derivation trees; the results were submitted to the proof checker and found correct, with the same conclusions.

Sixteen further proofs, transcribed from photographs of student work, were translated. Fourteen were well formed; all fourteen admitted positional translations and round-tripped exactly. The remaining two contained malformed justifications on the page—a rule cited with the wrong number of lines—and were rejected by the parser, as they should be.

We record the empirical finding for what it is worth: neither obstruction occurred in any student proof. They are not artefacts, as Section 4 shows, but they appear to be rare in practice.

8 What kind of equivalence is this?

It is tempting to summarise all of this by saying that Lemmon and Fitch notation are inter-translatable, and to leave it there. The results above suggest that the summary conceals more than it reports.

Translation from Fitch to Lemmon is total, canonical, and not injective (Theorem 4, Corollary 6). Translation from Lemmon to Fitch is total (Theorem 15) but not positional (Theorem 10). Composing in one order returns the original proof; composing in the other returns a normal form. In the vocabulary of structural relations between formalisms, Lemmon proofs are a *retract* of Fitch proofs, not an isomorphic copy of them.

The reason is that the two notations record different amounts. A Lemmon proof carries its dependency structure exactly, and carries no positional information at all. A Fitch proof carries its dependency structure only up to an upper bound, and carries positional information in addition—information which is, from the point of view of the derivation, arbitrary. Neither is a richer notation than the other. They are differently lossy.

That is a more interesting relationship than notational variance, and it is the sort of relationship that arises whenever two formalisms are said to say the same thing. The case is unusually clean: both formalisms are small, completely specified, and mechanically checkable, so the question of what survives translation can be settled rather than argued. We suggest it is worth having such a case in view when the same question is asked of theories, where it cannot.

References

- [1] Hans Halvorson. *How Logic Works: A User's Guide*. Princeton University Press, 2020.
- [2] E. J. Lemmon. *Beginning Logic*. Nelson, 1965.
- [3] Frederic B. Fitch. *Symbolic Logic: An Introduction*. Ronald Press, 1952.
- [4] Patrick Suppes. *Introduction to Logic*. Van Nostrand, 1957.